

NOTIFICATION EVENT CATALOGUE

Developer Handoff - Version 1.0

Companion document to:

IO SKY - Notification Center Functional Specification

Status: Functional Master / Developer Implementation & QA Handoff

Scope: All nine Notification Families

Purpose: Normative event registry, routing, permissions, action state, priority, delivery, quiet hours, digests, escalation, resolution, deduplication, idempotency, localisation, analytics and acceptance criteria.

Core rule: The right event - to the right authorised person - for the right resource - at the right time - with the right action - without exposing information they are not authorised to know.

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1. DOCUMENT PURPOSE

This document defines the normative Notification Event Catalogue for the IO SKY platform. It converts the higher-level Notification Center Functional Specification into a concrete registry of events that can be implemented, routed, tested, translated, delivered and monitored consistently.

This catalogue does not replace the existing Notification Center Functional Specification. The existing specification remains authoritative for Notification Center versus Audit Log separation; backend authorization; role, permission, scope and assignment rules; Client isolation; Technical Operator scopes; Admin Permission Presets; AI Scan review and publication separation; data minimisation; notification lifecycle; deep-link authorization; notification noise control; environment isolation; and Security Center access.

The existing access model remains: Role + Permission + Scope + Assignment/Ownership + Resource + Lifecycle State, with Environment added where relevant. Opening a notification never grants access by itself.

2. NOTIFICATION DESIGN PRINCIPLE

The Notification Center is an operational attention and action layer. It exists because an authorised recipient must act, should prepare, has received responsibility, needs awareness of a material lifecycle change, must resolve an exception, is approaching a meaningful deadline, or must address a security or operational risk.

- It is not a social feed.
- It is not an unrestricted activity log.
- It is not an Audit Log.
- It is not a backend process monitor.
- It is not an authorization system.
- It is not a record of every successful automated action.
- Routine healthy system activity must remain outside the Notification Center.

3. EVENT CLASSIFICATION

Every candidate system event must be classified as one of the following. Developers must not automatically convert every backend event into a Notification Center event.

- NOTIFY - Creates an authorised Notification Center event because human awareness, preparation or action is justified.
- AUDIT_ONLY - Creates traceability where appropriate but no Notification Center attention.
- NO_USER_EVENT - Routine internal operation requiring neither user attention nor a dedicated user-facing event.

4. GLOBAL EVENT IDENTIFIER RULE

Every event type in this catalogue must have a stable, language-independent identifier. Identifiers must not change when notification wording or translation changes. Examples include AI_SCAN_READY_FOR_REVIEW, PROJECT_DEADLINE_7D and SECURITY_ALERT_CRITICAL.

5. GLOBAL EVENT INSTANCE CONTRACT

Each generated logical notification event must support the following fields where applicable. The exact database schema is a development-team implementation decision.

Global Event Instance Contract

Field	Functional requirement
Event ID	Stable catalogue identifier
Logical event instance ID	Unique identifier for this occurrence
Family	One of the nine Notification Families
Resource type	Scan, Project, Call, Invoice, Integration, etc.
Resource ID	Authoritative resource reference
Organisation / Client ID	Required where tenant-related
Project ID	Where Project-scoped
Environment	Development / Staging / Production where relevant
Recipient ID	Authorised recipient
Created timestamp	Canonical server-generated time
Trigger timestamp	Time underlying business event occurred
Scheduled / due timestamp	Where applicable
Priority	P1-P4
Read state	Separate from operational resolution
Attention state	Actionable / Resolved / Superseded / Cancelled etc.
Template key	Stable localisation key
Deep-link target	Secure authorised destination
Dedupe reference	Logical deduplication identity
Delivery status	Per external channel
Audit relation	Where same underlying action requires audit evidence

6. AUTHORIZATION RULE

Notification routing must be calculated server-side. Frontend hiding alone is never sufficient. Every relevant event must evaluate Role + Permission + Scope + Assignment/Ownership + Resource + Lifecycle State + Environment where applicable.

A notification received before permission revocation must never preserve resource access after that permission is removed. Every deep link must re-check current backend authorization.

7. PRIORITY MODEL

P1 - Critical. Immediate attention required. Examples: critical security incident, critical platform outage, serious access compromise, or business-critical failure requiring immediate human intervention.

P2 - High. Important operational action is required or a significant deadline/risk is approaching. Examples: AI Scan Ready for Review, AI Scan ready to Publish, failed payment, project deadline tomorrow, unresolved blocker threatening delivery.

P3 - Normal. Relevant preparation or operational lifecycle information. Examples: Discovery Call in 24 hours, Project assigned, requested Client information received, invoice due in seven days.

P4 - Informational. Useful information that does not currently require urgent action. P4 should be used sparingly.

8. ACTION REQUIRED MODEL

Each notification must explicitly carry one of four functional attention states.

- ACTION_REQUIRED - Recipient must perform an action.
- PREPARATION_REQUIRED - Recipient should prepare for an upcoming event.
- AWARENESS - No immediate action required but material awareness is justified.
- RESOLUTION_NOTICE - A previously actionable condition has materially resolved.

9. READ STATE AND OPERATIONAL STATE

Read/unread is independent from action/resolution. READ + ACTION_REQUIRED is valid. UNREAD + RESOLVED is also valid. Opening a notification does not automatically complete the underlying work.

10. IN-APP DELIVERY

Unless an event is explicitly external-only, every authorised internal platform notification in this catalogue generates an in-app Notification Center event. In-app notification creation is not suppressed by quiet hours.

11. EMAIL DELIVERY VALUES

Each event uses one of the following delivery semantics. Mandatory delivery does not override resource authorization.

- MANDATORY - Send email to authorised recipient when configured email exists.
- IMMEDIATE - Delivery should occur immediately and must not wait for a digest.
- OPTIONAL - May be controlled through permitted notification preferences.
- DIGEST_ELIGIBLE - May be grouped into an authorised digest.
- NONE - No external email notification is required.

12. QUIET HOURS

Quiet hours primarily govern external interruption, not existence of the underlying in-app notification.

RESPECT means external notification may wait until quiet hours end. BYPASS means external notification must be allowed to send immediately. NOT_APPLICABLE means no external delivery is used.

Quiet hours must never remove the in-app event itself. P1 Critical Security and genuinely critical Platform incidents may bypass quiet hours. Ordinary operational notifications respect quiet hours.

13. DIGEST RULE

Events marked DIGEST = NO must not be delayed into a digest. Events marked DIGEST = YES may be grouped where doing so does not reduce operational effectiveness. Critical events, imminent deadlines and action-required AI Scan review/publication events are not digest eligible.

14. DELIVERY RETRY

External delivery failure is a delivery problem, not a new business event. For example, AI_SCAN_READY_FOR_REVIEW is generated once. If email delivery fails three times, the platform still has one logical event.

Retry architecture, provider and backoff algorithm remain developer decisions, but required delivery must not be silently lost; retries must be observable; retries must not duplicate the logical event; and persistent delivery failure must be detectable.

15. IDEMPOTENCY

Event processing must be idempotent. Processing the same logical event more than once must not create duplicate Notification Center rows, duplicate emails, duplicate Client publication or duplicate irreversible business actions.

This is especially important for AI Scan Publish: one successful logical Publish operation must create one published Client-visible result.

16. DEDUPLICATION

The default logical dedupe basis is Event Type + Resource + Recipient + Relevant Lifecycle Occurrence. A genuinely new actionable occurrence must not be accidentally suppressed. A Scan may legitimately become Ready for Review again after correction/reprocessing; that is a new lifecycle occurrence.

17. SUPERSESSION

Notifications become non-actionable when their underlying condition is completed, cancelled, resolved, replaced, rescheduled, superseded or otherwise rendered obsolete. Historical visibility may remain where useful, but obsolete notifications must not continue to appear as unresolved attention.

18. TIMEZONE AND SCHEDULING RULE

All scheduled reminders must be calculated from authoritative server-side timestamps. The platform must not rely on the user's device clock.

Technical storage implementation remains a development decision, but the implementation must support reliable timezone conversion and daylight-saving behavior. Display must use the relevant

configured platform/user/resource timezone. Reminder generation must not produce duplicate or missing reminders due solely to DST/timezone changes.

19. LOCALISATION

Event semantics and Event IDs are language-independent. Each notification type must use a stable translation/template key, for example notification.ai_scan.ready_for_review.title.

Copy may be rendered according to the recipient's configured platform language. Client-facing notifications must use the configured Client-facing language where supported. Translation may never alter authorization, priority, event meaning, underlying state, ownership or action required.

20. DELIVERY ANALYTICS

The system must support operational visibility into event generated, recipient resolved, notification created, channel delivery attempted, delivery success, delivery failure, retry count/status, read timestamp, underlying action resolution, escalation and duplicate suppression where relevant. Sensitive payloads must not be unnecessarily copied into analytics.

21. NOTIFICATION FAMILY 1 - DISCOVERY CALLS

DC-01 - DISCOVERY_CALL_REQUESTED

Classification	NOTIFY
Trigger	New valid Discovery Call request created.
Recipients	Super Admin; Admin with effective Discovery Call permission and relevant responsibility/assignment.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Unauthenticated prospect does not require an in-app object; external confirmation uses approved scheduling/transactional channel.
Action type	ACTION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	If request remains unassigned/unhandled beyond an approved configurable operational threshold.
Resolution / supersession	Request handled, scheduled, cancelled or otherwise closed.
Deduplication	Discovery Call request ID + recipient
Deep link	Authorised Discovery Call record
Sensitive content	Minimal prospect/company context only.
Audit relation	As applicable

DC-02 - DISCOVERY_CALL_SCHEDULED

Classification	NOTIFY
Trigger	Discovery Call receives a confirmed scheduled date/time.
Recipients	Current assigned responsible internal user; authenticated Client where applicable; external prospect via approved transactional scheduling channel.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes where authenticated; external prospect via approved transactional channel.
Action type	PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	MANDATORY
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Call cancelled, completed or rescheduled.
Deduplication	Call ID + scheduled occurrence + recipient
Deep link	Authorised Discovery Call record
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

DC-03 - DISCOVERY_CALL_RESCHEDULED

Classification	NOTIFY
Trigger	Confirmed scheduled date/time changes.
Recipients	Current assigned responsible user(s) and relevant authenticated/external participant.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes where relevant.
Action type	AWARENESS / PREPARATION_REQUIRED
Priority	P2 when close to original/new time; otherwise P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	New schedule becomes authoritative.
Deduplication	Call ID + reschedule version + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	All reminders belonging to the old scheduled time must be cancelled/superseded.

DC-04 - DISCOVERY_CALL_CANCELLED

Classification	NOTIFY
Trigger	Scheduled Discovery Call is cancelled.
Recipients	Current responsible users and relevant participant.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes where relevant.
Action type	AWARENESS
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Cancellation is final unless Call is explicitly reopened/rebooked.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	All future 24h and 1h reminders must be cancelled.

DC-05 - DISCOVERY_CALL_UPCOMING_24H

Classification	NOTIFY
Trigger	24 hours before authoritative scheduled start, only while Call remains active.
Recipients	Current assigned authorised internal responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Call cancelled, rescheduled, completed, or 1h reminder becomes the current imminent reminder.
Deduplication	Call ID + scheduled occurrence + 24H + recipient
Deep link	Discovery Call with intake/prospect context
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Generate only if Call remains scheduled, recipient remains assigned and recipient remains authorised.

DC-06 - DISCOVERY_CALL_UPCOMING_1H

Classification	NOTIFY
Trigger	One hour before authoritative scheduled start, only while Call remains active.
Recipients	Current assigned authorised responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	PREPARATION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Call starts/completes/cancels/reschedules.
Deduplication	Call ID + scheduled occurrence + 1H + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

DC-07 - DISCOVERY_CALL_RETRY_DUE

Classification	NOTIFY
Trigger	Explicit retry/next-attempt timestamp is reached.
Recipients	Current assigned responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3 based on urgency
In-app	Yes
Email / external delivery	IMMEDIATE for P2; OPTIONAL for P3
Quiet hours	RESPECT
Digest	NO
Escalation	Based on configured next-action policy if repeatedly unhandled.
Resolution / supersession	Retry completed, cancelled or superseded by new follow-up.
Deduplication	Call ID + retry timestamp/version + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	No arbitrary retry schedule may be invented; this follows an explicit business retry timestamp.

DC-08 - DISCOVERY_CALL_FOLLOW_UP_DUE

Classification	NOTIFY
Trigger	Explicit follow-up/next-action deadline is reached.
Recipients	Assigned responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Follow-up completed, cancelled or superseded.
Deduplication	Call ID + follow-up occurrence + recipient
Deep link	Discovery Call/prospect record
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

DC-09 - DISCOVERY_CALL_INFORMATION_UPDATED

Classification	NOTIFY
Trigger	Material new prospect/Client information relevant to an upcoming or active Call is added or changed.
Recipients	Current responsible user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES
Escalation	None unless explicitly stated
Resolution / supersession	Informational; no persistent action state unless underlying action is separately created.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Do not generate for trivial field edits, formatting changes or autosave.

DC-10 - DISCOVERY_CALL_ACTION_REQUIRED

Classification	NOTIFY
Trigger	A genuine Discovery Call action exists and no more specific Discovery Call event applies.
Recipients	Clearly responsible authorised user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where time-sensitive
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Underlying action completed/cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Restricted fallback only; must never become a generic dumping ground.

22. DISCOVERY CALL AUDIT-ONLY / NO-EVENT ACTIVITY

- Ordinary record viewing.
- Routine note saves.
- Autosave.
- Routine backend activity.
- Harmless minor field changes.
- Healthy scheduling processing.

23. NOTIFICATION FAMILY 2 - AI SCANS

Applies to Operations Scan, Cyber Scan and Elite Scan. Mandatory lifecycle: Assessment → Analysis → Human Review → Approval → Publish → Client Portal. Clients must never see internal AI Scan analysis, unpublished findings or review states before authorised publication.

AS-01 - AI_SCAN_CREATED

Classification	NOTIFY
Trigger	Valid Scan instance is created/purchased.
Recipients	Relevant authorised AI Scan Operations user(s) where operational awareness is required.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Client may receive purchase confirmation through appropriate Client-facing channel.
Action type	AWARENESS
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL internally; Client purchase confirmation transactional as appropriate
Quiet hours	RESPECT
Digest	YES internally where no action is required
Escalation	None unless explicitly stated
Resolution / supersession	Scan progresses or is otherwise closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AS-02 - AI_SCAN_INFORMATION_REQUIRED

Classification	NOTIFY
Trigger	Required Client input is missing and Scan progress is blocked or cannot correctly continue.
Recipients	Correct Client; authorised assigned Scan Operations responsible user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes - only describe what the Client needs to provide.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	If information remains outstanding and delivery window is threatened, escalate internally to responsible Scan Operations + Super Admin.
Resolution / supersession	Required information supplied or requirement explicitly cancelled.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Client Scan input/request view or internal Scan operations view
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AS-03 - AI_SCAN_SUBMITTED

Classification	NOTIFY
Trigger	Client successfully submits completed assessment.
Recipients	Assigned authorised Scan Operations users; Super Admin where appropriate for oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes - submission confirmation.
Action type	AWARENESS
Priority	P3
In-app	Yes
Email / external delivery	Client confirmation MANDATORY/transactional; internal OPTIONAL
Quiet hours	RESPECT
Digest	YES internally where no immediate action is required
Escalation	None unless explicitly stated
Resolution / supersession	Analysis begins/progresses.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AS-04 - AI_SCAN_ANALYSIS_EXCEPTION

Classification	NOTIFY
Trigger	Analysis materially fails/cannot continue and requires human intervention.
Recipients	Authorised Scan Operations responsible user; relevant authorised technical operator where technically assigned; Super Admin when impact is material.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No internal technical detail.
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT unless business-critical
Digest	NO
Escalation	Escalate to Super Admin when unresolved and Scan delivery becomes at risk.
Resolution / supersession	Analysis successfully resumes/completes or Scan is operationally closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	No substantive findings, secrets or unnecessary technical internals.
Audit relation	As applicable

AS-05 - AI_SCAN_READY_FOR_REVIEW

Classification	NOTIFY
Trigger	Analysis successfully completes and reviewable report output is available.
Recipients	Super Admin; Admin explicitly authorised with scan.review and access/assignment; Developer only when explicitly assigned as Reviewer for this specific Scan and given required review authorization.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No. Never Client-visible before authorised publication.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	Deadline-aware; AI_SCAN_DELIVERY_AT_RISK when configured delivery window is threatened.
Resolution / supersession	Review completed; reviewer reassigned where old reviewer is no longer responsible; Scan returned to analysis/correction; or Scan cancelled/superseded.
Deduplication	Scan ID + review lifecycle version + recipient
Deep link	Internal Scan Review Workspace
Sensitive content	Preview must not contain substantive findings.
Audit relation	As applicable
Implementation notes	Ordinary Developer, Technical Operator and Client are not recipients merely by role.

AS-06 - AI_SCAN_REVIEW_ASSIGNED

Classification	NOTIFY
Trigger	Specific reviewer is assigned or reassigned to a Scan.
Recipients	Newly assigned authorised Reviewer.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Review completed, reassigned or cancelled.
Deduplication	Scan ID + assignment version + reviewer
Deep link	Assigned Scan Review Workspace
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Reviewer assignment must not provide access to unrelated Scans.

AS-07 - AI_SCAN_REVIEW_REMINDER

Classification	NOTIFY
Trigger	Review remains unresolved and an approved deadline-based reminder point is reached.
Recipients	Current assigned Reviewer.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	If delivery risk threshold is reached, generate AI_SCAN_DELIVERY_AT_RISK.
Resolution / supersession	Review completed/reassigned/cancelled/superseded.
Deduplication	Scan ID + reminder threshold + current reviewer
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AS-08 - AI_SCAN_CHANGES_REQUIRED

Classification	NOTIFY
Trigger	Reviewer explicitly determines output requires correction/reprocessing before approval.
Recipients	Appropriate authorised person/team responsible for correction or analysis.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Revised version becomes available for review.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Review decision must be traceable.

AS-09 - AI_SCAN_REVIEW_APPROVED

Classification	NOTIFY
Trigger	Authorised Reviewer clicks Approve; reviewed report version enters Approved / Ready for Publication state.
Recipients	Super Admin; Admin with explicit scan.publish permission AND access to that Scan.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No until publication succeeds.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Successfully published; approval revoked through authorised correction workflow; or Scan superseded.
Deduplication	Scan ID + approved report version + publisher recipient
Deep link	Approved Scan / Publish Workspace
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Approval event must be traceable.
Implementation notes	Notification context should minimally show Client, Scan type, Approved status, reviewer identity, approval timestamp, report language and approved version identifier. Reviewer authority alone does not grant Publish.

24. AI SCAN REVIEW / PUBLISH PERMISSION MATRIX

Review / Publish Permission Matrix

Actor	View assigned review	Approve review	Publish
Super Admin	Yes	Yes	Yes
Admin + scan.review + required resource access	Yes	Yes	No unless separately granted scan.publish
Admin + scan.publish but no review permission	Publisher view as required	No review approval right	Yes once approved
Developer explicitly assigned as Scan Reviewer	Yes	Yes	No by default
Developer without Scan Review assignment	No	No	No
Technical Operator by role alone	No	No	No
Client	No	No	No

25. AI SCAN ONE-ACTION PUBLISH

Publication must be operationally simple. The authorised Publisher opens the approved Scan and receives one clear action: Publish. No unnecessary intermediate workflow is required.

Before accepting Publish, backend validation must confirm that the Scan exists; the lifecycle permits publication; an approved report version exists; the current user has scan.publish; the user has resource access; Client ownership is valid; the intended Client account is valid; the required Client-facing report language/version exists; and the version being published is the approved version.

After one valid Publish action, the system must persist publication state; bind publication to the correct Client; make the approved version Client-visible; place it in the correct Client Portal; expose the correct Client-selected report language; keep internal review/draft data private; record publication audit evidence; and generate the Client availability notification only after success.

AS-10 - AI_SCAN_PUBLICATION_FAILED

Classification	NOTIFY
Trigger	Authorised Publish attempt fails before the correct approved report has become successfully available in the correct Client Portal.
Recipients	Attempting Publisher; Super Admin.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No notification.
Action type	ACTION_REQUIRED
Priority	P1/P2 depending on delivery impact
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	P1 immediate; P2 according to delivery risk.
Resolution / supersession	Successful publication or deliberate authorised operational resolution.
Deduplication	Scan ID + publication failure lifecycle; technical retries must not flood users.
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Failed privileged publication attempt must be traceable.
Implementation notes	The system must not produce AI_SCAN_PUBLISHED or a Client availability notification unless publication success has been verified.

AS-11 - AI_SCAN_PUBLISHED

Classification	NOTIFY
Trigger	Authorised publication completed successfully and correct approved report is available in correct Client Portal.
Recipients	Correct Client; relevant authorised internal users where useful.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	AWARENESS / View report
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE to Client
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Informational availability event; report remains available according to portal lifecycle.
Deduplication	Scan ID + published approved version + recipient
Deep link	Correct published Client Portal report
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Publisher, timestamp, version, Client and publication state must be traceable.
Implementation notes	Client receives only the published version. Client-facing notification uses configured Client language where supported.

AS-12 - AI_SCAN_DELIVERY_AT_RISK

Classification	NOTIFY
Trigger	Remaining configured delivery time is materially threatened while required review, approval or publication remains outstanding.
Recipients	Current responsible reviewer/publisher as applicable; Super Admin.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	This event is itself an escalation state.
Resolution / supersession	Risk cleared through progression or Scan delivery completed.
Deduplication	Scan ID + risk lifecycle/version + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AS-13 - AI_SCAN_DELIVERY_OVERDUE

Classification	NOTIFY
Trigger	Applicable committed Scan delivery deadline passes while Client-facing delivery remains incomplete.
Recipients	Super Admin; current responsible authorised internal user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No internal workflow details are exposed.
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS only if severity warrants; otherwise RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Delivery completed or authorised exception resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

26. AI SCAN LANGUAGE REQUIREMENT

The Client-selected report language must be part of the publication validation context. If a required Client-facing report language/version is missing, Publish must not falsely succeed. Do not publish another language with the assumption that translation can happen later.

27. NOTIFICATION FAMILY 3 - CLIENTS & ACCOUNTS

CA-01 - CLIENT_ACCOUNT_CREATED

Classification	NOTIFY
Trigger	Client account/organisation created and relevant operational awareness is required.
Recipients	Assigned authorised Client-management users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Client may receive onboarding/account-access communication separately.
Action type	AWARENESS / ACTION_REQUIRED depending on onboarding
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL internally; onboarding communication as required externally
Quiet hours	RESPECT
Digest	YES internally if not action-required
Escalation	None unless explicitly stated
Resolution / supersession	Onboarding/assignment completed or account state superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

CA-02 - CLIENT_INFORMATION_REQUIRED

Classification	NOTIFY
Trigger	Authorised internal user explicitly requests Client information/documentation.
Recipients	Correct Client; assigned responsible internal user where tracking is useful.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED for Client
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE to Client
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Requested information received or request cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

CA-03 - CLIENT_INFORMATION_SUBMITTED

Classification	NOTIFY
Trigger	Client supplies requested information/documentation.
Recipients	Current assigned responsible internal user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED or AWARENESS depending on workflow
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL; IMMEDIATE where work is blocked awaiting this input
Quiet hours	RESPECT
Digest	YES only if no action is urgent
Escalation	None unless explicitly stated
Resolution / supersession	Underlying request/workflow progresses or is closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

CA-04 - CLIENT_MATERIAL_INFORMATION_UPDATED

Classification	NOTIFY
Trigger	Material Client information changes in a way requiring authorised awareness/action.
Recipients	Assigned relevant user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES
Escalation	None unless explicitly stated
Resolution / supersession	Informational unless a separate action is created.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	No event for routine profile edits without operational significance.

CA-05 - CLIENT_ACCOUNT_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Genuine Client-account action exists and no more specific event applies.
Recipients	Correct Client and/or assigned authorised internal user dictated by underlying action.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where deadline exists
Quiet hours	RESPECT
Digest	NO when action deadline exists
Escalation	None unless explicitly stated
Resolution / supersession	Underlying action completed/cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Strict fallback only.

CA-06 - CLIENT_ACCOUNT_SECURITY_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Client must take account-level security action.
Recipients	Affected Client; authorised security oversight only where necessary.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes, but no internal Security Center details.
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; otherwise RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Required security action completed or authorised resolution.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal safe Client-facing security context only.
Audit relation	As applicable

28. CLIENT / TENANT ISOLATION

Client A must never receive Client B's notification data. Admin access to one Client must not imply access to all Clients. Client notifications may only expose the Client's own authorised resources.

29. NOTIFICATION FAMILY 4 - PROJECTS & DELIVERY

PD-01 - PROJECT_ASSIGNED

Classification	NOTIFY
Trigger	User receives responsibility for Project/resource.
Recipients	Newly assigned authorised user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / ACTION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Assignment completed, removed or reassigned.
Deduplication	Project ID + assignment version + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-02 - PROJECT_REASSIGNED

Classification	NOTIFY
Trigger	Project responsibility changes.
Recipients	New responsible user; previous user only where operational handoff awareness is necessary.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	New assignment becomes authoritative.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Future notifications follow current assignment.

PD-03 - PROJECT_MILESTONE_UPDATED

Classification	NOTIFY
Trigger	Material milestone state/date/scope changes.
Recipients	Assigned authorised Project participants whose work/responsibility is affected.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES if not time-critical
Escalation	None unless explicitly stated
Resolution / supersession	Informational or superseded by later milestone state.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	No event for cosmetic/minor edits.

PD-04 - PROJECT_DEADLINE_7D

Classification	NOTIFY
Trigger	Active incomplete milestone/deliverable reaches seven days before deadline.
Recipients	Responsible Implementer/owner + relevant Project/Delivery Manager.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Completed, cancelled or superseded.
Deduplication	Project/milestone ID + deadline version + 7D + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-05 - PROJECT_DEADLINE_3D

Classification	NOTIFY
Trigger	Active incomplete work reaches three days before deadline.
Recipients	Current responsible users + appropriate delivery oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED / PREPARATION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Completed, cancelled or superseded.
Deduplication	Project/milestone ID + deadline version + 3D + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-06 - PROJECT_DEADLINE_24H

Classification	NOTIFY
Trigger	Active incomplete milestone/deliverable reaches 24 hours before deadline.
Recipients	Responsible user + appropriate Project/Delivery oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Completed, cancelled or superseded.
Deduplication	Project/milestone ID + deadline version + 24H + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	This is a final reminder, not the first warning.

PD-07 - PROJECT_DEADLINE_MISSED

Classification	NOTIFY
Trigger	Deadline passes while required work remains incomplete.
Recipients	Assigned responsible user; Project Manager/Delivery Manager with effective authorization; Super Admin where escalation policy requires.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	Escalate according to authorised Project/Delivery responsibility and configured policy.
Resolution / supersession	Work completed or deadline/status formally superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-08 - PROJECT_REVIEW_REQUESTED

Classification	NOTIFY
Trigger	Work formally enters Review Requested state.
Recipients	Explicitly assigned Reviewer.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3 depending on deadline
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Review completed/reassigned/cancelled.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Developer role alone does not trigger review notification; actual Reviewer assignment must exist.

PD-09 - PROJECT_REVIEWER_ASSIGNED

Classification	NOTIFY
Trigger	Reviewer is assigned/reassigned.
Recipients	Newly assigned Reviewer.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Review completed/reassigned.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-10 - PROJECT_IMPLEMENTATION_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Concrete implementation action is assigned.
Recipients	Assigned Implementer.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where deadline/action is material
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Action completed/cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-11 - PROJECT_REVIEW_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Concrete review action is required.
Recipients	Assigned Reviewer only.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Review action completed/reassigned/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-12 - PROJECT_BLOCKER_CREATED

Classification	NOTIFY
Trigger	Material blocker exists that prevents or threatens delivery.
Recipients	Responsible assignee; relevant Project/Delivery oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2; P1 only if business-critical
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; otherwise RESPECT
Digest	NO
Escalation	If unresolved at configured blocker escalation threshold or threatens Project deadline, escalate to higher delivery oversight/Super Admin as authorised.
Resolution / supersession	Blocker marked resolved or superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-13 - PROJECT_BLOCKER_RESOLVED

Classification	NOTIFY
Trigger	Active material blocker is resolved.
Recipients	Users whose next action/dependency changes because of resolution.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	RESOLUTION_NOTICE
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES if non-urgent
Escalation	None unless explicitly stated
Resolution / supersession	Resolution notice is informational; related blocker notification must clear.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Related blocker notification must no longer appear unresolved.

PD-14 - PROJECT_DEPENDENCY_ACTION_REQUIRED

Classification	NOTIFY
Trigger	External/internal dependency requires concrete action.
Recipients	Assigned dependency owner + relevant Project oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where delivery is affected
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Dependency satisfied, cancelled or superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-15 - PROJECT_CLIENT_INFORMATION_REQUIRED

Classification	NOTIFY
Trigger	Project requires information/documentation from Client.
Recipients	Correct Client; assigned Project responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE to Client
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Information received or request cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-16 - PROJECT_CLIENT_INFORMATION_RECEIVED

Classification	NOTIFY
Trigger	Client provides requested Project information/documentation.
Recipients	Current assigned internal responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / ACTION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL; IMMEDIATE where work was blocked
Quiet hours	RESPECT
Digest	YES only if no immediate action is required
Escalation	None unless explicitly stated
Resolution / supersession	Underlying Project work progresses.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PD-17 - PROJECT_DELIVERY_STATE_CHANGED

Classification	NOTIFY
Trigger	Material delivery lifecycle transition such as ready for review, approved, delivery ready, delivered, or paused where attention is required.
Recipients	Only authorised stakeholders affected by the transition.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / ACTION_REQUIRED depending on state
Priority	P3 or P2
In-app	Yes
Email / external delivery	Context dependent
Quiet hours	RESPECT
Digest	YES only for non-urgent informational transitions
Escalation	None unless explicitly stated
Resolution / supersession	Superseded by next authoritative delivery state.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

30. NOTIFICATION FAMILY 5 - PAYMENTS & BILLING

Billing notifications require effective Billing authorization. Admin role alone is insufficient.

PB-01 - INVOICE_CREATED_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Invoice creation requires human attention.
Recipients	Authorised Billing user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES
Escalation	None unless explicitly stated
Resolution / supersession	Invoice issue/action resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Routine successful automated invoice generation does not need a Notification Center event.

PB-02 - PAYMENT_DUE_7D

Classification	NOTIFY
Trigger	Invoice remains unpaid seven days before due date.
Recipients	Correct Client billing recipient; authorised internal Billing responsible user where operationally relevant.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	PREPARATION_REQUIRED
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Payment completed, invoice cancelled or superseded.
Deduplication	Invoice ID + due-date version + 7D + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PB-03 - PAYMENT_DUE_1D

Classification	NOTIFY
Trigger	Invoice remains unpaid one day before due date.
Recipients	Correct Client billing recipient + authorised Billing responsible user as appropriate.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Payment completed, invoice cancelled or superseded.
Deduplication	Invoice ID + due-date version + 1D + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PB-04 - PAYMENT_DUE_TODAY

Classification	NOTIFY
Trigger	Invoice remains unpaid on due date.
Recipients	Correct Client billing recipient + authorised Billing responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Payment completed, invoice cancelled or superseded.
Deduplication	Invoice ID + due-date version + DUE + recipient
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PB-05 - PAYMENT_OVERDUE

Classification	NOTIFY
Trigger	Invoice remains unpaid after due date.
Recipients	Client billing recipient; authorised internal Billing responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	Further commercial/legal action is outside this catalogue and must follow approved IO SKY contractual/legal rules.
Resolution / supersession	Payment completed, invoice legally/operationally superseded or authorised closure.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PB-06 - PAYMENT_FAILED

Classification	NOTIFY
Trigger	Payment attempt fails and human action is required.
Recipients	Correct Client; authorised Billing responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes.
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Successful payment or authorised alternative resolution.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Do not expose sensitive processor/security details.
Audit relation	As applicable

PB-07 - PAYMENT_COMPLETED

Classification	NOTIFY
Trigger	Payment successfully completed.
Recipients	Client confirmation; relevant Billing user where operationally useful.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	Yes as appropriate.
Action type	AWARENESS
Priority	P4/P3
In-app	Yes
Email / external delivery	Client receipt/transactional as applicable; internal OPTIONAL
Quiet hours	RESPECT
Digest	YES internally
Escalation	None unless explicitly stated
Resolution / supersession	Informational.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Generate Notification Center event only where completion is operationally meaningful. Payment state itself must always update.

PB-08 - BILLING_ACTION_REQUIRED

Classification	NOTIFY
Trigger	A real Billing action exists and no more specific event applies.
Recipients	Authorised Billing responsible user and/or correct Client where underlying action requires.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where time-sensitive
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Underlying action completed/cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Strict fallback only.

31. NO REFUND / CHARGEBACK NOTIFICATION WORKFLOW

No refund or chargeback workflow or Notification Family is introduced through this specification.

32. NOTIFICATION FAMILY 6 - AI OPERATIONS

AI Operations concerns technical/operational AI-process health, not the AI Scan business lifecycle itself.

AO-01 - AI_OPERATION_FAILED

Classification	NOTIFY
Trigger	AI process cannot continue successfully without intervention.
Recipients	Authorised assigned AI/technical responsible users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P1 depending on impact
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Process successfully recovered/restarted or intentionally terminated.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AO-02 - AI_HUMAN_INTERVENTION_REQUIRED

Classification	NOTIFY
Trigger	AI workflow explicitly requires authorised human decision/action.
Recipients	Assigned authorised responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Human intervention completed or workflow superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AO-03 - AI_PROCESSING_EXCEPTION

Classification	NOTIFY
Trigger	Material AI processing exception requires investigation.
Recipients	Assigned operational/technical responsible users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Exception resolved/recovered/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Repeated backend retries of the same exception must not flood the Notification Center.

AO-04 - AI_OPERATION_THRESHOLD_EXCEEDED

Classification	NOTIFY
Trigger	Explicitly approved actionable AI operational threshold is exceeded.
Recipients	Authorised responsible user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3 depending on threshold
In-app	Yes
Email / external delivery	IMMEDIATE where action required
Quiet hours	RESPECT
Digest	NO if action required
Escalation	None unless explicitly stated
Resolution / supersession	Threshold returns to acceptable state or authorised resolution.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Thresholds must be explicitly configured rather than inferred.

AO-05 - AI_SERVICE_UNAVAILABLE

Classification	NOTIFY
Trigger	Required AI service/provider becomes unavailable with material platform impact.
Recipients	Assigned authorised operational/technical users + Super Admin for material outage.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Service restored or authorised workaround/closure.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

AO-06 - AI_OPERATION_RECOVERED

Classification	NOTIFY
Trigger	Previously material AI incident recovers.
Recipients	Users actively affected/responsible.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	RESOLUTION_NOTICE
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL
Quiet hours	RESPECT
Digest	YES where no immediate resumed action is required
Escalation	None unless explicitly stated
Resolution / supersession	Resolution notice is informational; prior incident attention clears.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

33. AI OPERATIONS NOISE EXCLUSIONS

- Every successful AI request.
- Successful model response.
- Ordinary processing stage.
- Normal queue progress.
- Successful automated retry.
- Ordinary API/token usage.
- Healthy AI provider status.

34. NOTIFICATION FAMILY 7 - PLATFORM & INTEGRATIONS

PI-01 - INTEGRATION_DISCONNECTED

Classification	NOTIFY
Trigger	Required integration loses working connection.
Recipients	Authorised owner/operator of affected integration.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT unless critical Production impact
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Integration restored or intentionally disconnected.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-02 - INTEGRATION_AUTH_EXPIRED

Classification	NOTIFY
Trigger	Integration authentication/authorization expires and human action is required.
Recipients	Authorised integration owner/operator.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Authentication restored or integration intentionally disabled.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-03 - INTEGRATION_FAILURE

Classification	NOTIFY
Trigger	Integration fails materially and operation is affected.
Recipients	Appropriate authorised technical/operational owner.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P1
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; otherwise RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Integration recovers or issue is resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-04 - EXTERNAL_SERVICE_UNAVAILABLE

Classification	NOTIFY
Trigger	External service outage materially impacts platform capability.
Recipients	Authorised responsible technical/operational users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	Impact-based P1/P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE for P1/P2
Quiet hours	P1 BYPASS; otherwise RESPECT
Digest	NO when impact is active
Escalation	None unless explicitly stated
Resolution / supersession	Service recovers or authorised workaround/closure.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-05 - PLATFORM_COMPONENT_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Platform component requires authorised intervention.
Recipients	Authorised component owner/operator.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT unless separately P1
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Required intervention completed or issue superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-06 - PLATFORM_CONFIGURATION_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Misconfiguration materially affects operation and requires action.
Recipients	Authorised responsible user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where blocking
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Configuration corrected or issue superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Ordinary successful configuration changes are Audit Only; privileged changes should be traceable.

PI-07 - PLATFORM_INCIDENT_CRITICAL

Classification	NOTIFY
Trigger	Critical platform incident requires immediate intervention.
Recipients	Appropriate authorised technical responsible users; Super Admin.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P1
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	BYPASS
Digest	NEVER
Escalation	Immediate according to authorised operational incident chain.
Resolution / supersession	Incident confirmed resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

PI-08 - PLATFORM_SERVICE_RECOVERED

Classification	NOTIFY
Trigger	Previously material outage is resolved.
Recipients	Affected responsible users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	RESOLUTION_NOTICE
Priority	P3
In-app	Yes
Email / external delivery	OPTIONAL / IMMEDIATE if teams were actively responding
Quiet hours	RESPECT
Digest	YES where no immediate restart action is needed
Escalation	None unless explicitly stated
Resolution / supersession	Resolution notice informational; related incident attention clears.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

35. PLATFORM NOISE EXCLUSIONS

- Routine health checks.
- Successful background jobs.
- Ordinary sync success.
- Healthy integration heartbeat.
- Normal database operation.
- Every successful deployment.
- Routine internal processing.

36. NOTIFICATION FAMILY 8 - SECURITY & ACCESS

Security Center visibility cannot be inferred from Admin or Technical Operator role alone. Explicit Security authorization/scope is required.

SA-01 - SECURITY_ALERT_CRITICAL

Classification	NOTIFY
Trigger	Confirmed critical security condition requires immediate human attention.
Recipients	Super Admin; explicitly authorised Security user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P1
In-app	Yes
Email / external delivery	IMMEDIATE / MANDATORY
Quiet hours	BYPASS
Digest	NEVER
Escalation	Immediate.
Resolution / supersession	Security incident formally resolved/closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Authorised Security Center record
Sensitive content	Minimal preview only; sensitive details remain in Security Center.
Audit relation	Mandatory traceability.

SA-02 - SECURITY_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Material security action is required but incident is not necessarily P1.
Recipients	Explicitly authorised Security user(s).
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT unless severity escalates
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Security action completed or issue superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As appropriate for security action.

SA-03 - PRIVILEGED_ACCESS_GRANTED

Classification	NOTIFY
Trigger	Privileged platform/security access is granted.
Recipients	Super Admin and authorised governance/security oversight where appropriate.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / oversight
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE for sensitive privilege
Quiet hours	RESPECT
Digest	NO for high privilege
Escalation	None unless explicitly stated
Resolution / supersession	Informational/oversight state completed or superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Mandatory traceability.

SA-04 - PRIVILEGED_ACCESS_REVOKED

Classification	NOTIFY
Trigger	Privileged access is revoked.
Recipients	Appropriate authorised oversight users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS
Priority	P2/P3
In-app	Yes
Email / external delivery	Context dependent
Quiet hours	RESPECT
Digest	Context dependent
Escalation	None unless explicitly stated
Resolution / supersession	Informational.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Mandatory.
Implementation notes	Existing stale notifications must not retain resource access.

SA-05 - SIGNIFICANT_PERMISSION_CHANGE

Classification	NOTIFY
Trigger	Material permission change warrants security/governance awareness.
Recipients	Authorised oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / ACTION_REQUIRED where review is needed
Priority	P2/P3
In-app	Yes
Email / external delivery	IMMEDIATE where high privilege
Quiet hours	RESPECT
Digest	Context dependent
Escalation	None unless explicitly stated
Resolution / supersession	Oversight completed or change superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Mandatory.
Implementation notes	Routine low-risk assignment changes do not belong here.

SA-06 - SECURITY_CENTER_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Security Center record requires concrete authorised action.
Recipients	Only users with explicit Security Center authorization.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT unless escalated to P1
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Security Center action completed/resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

SA-07 - SECURITY_CONFIGURATION_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Security configuration requires intervention.
Recipients	Explicit Security-authorized users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Configuration corrected/resolved.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Privileged changes/resolution must be traceable.

SA-08 - SUSPICIOUS_ACCESS_EVENT

Classification	NOTIFY
Trigger	Security controls classify access activity as sufficiently suspicious to justify human attention.
Recipients	Explicit Security-authorised users; affected user only when appropriate user-facing security action is required.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Investigation/action resolved or event dismissed by authorised workflow.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal safe information; no unnecessary exploit or credential details.
Audit relation	As applicable
Implementation notes	Ordinary successful login does not automatically generate this event.

SA-09 - ACCOUNT_SECURITY_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Specific user must perform a security action.
Recipients	Affected authorised account holder.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	May be Client-facing when the affected account is a Client.
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Required account security action completed or authorised resolution.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal safe information.
Audit relation	As applicable

SA-10 - SECURITY_INCIDENT_RESOLVED

Classification	NOTIFY
Trigger	Active security incident is formally resolved.
Recipients	Authorised stakeholders who were responsible/affected.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	RESOLUTION_NOTICE
Priority	P3
In-app	Yes
Email / external delivery	IMMEDIATE where incident response was active
Quiet hours	RESPECT
Digest	YES if non-urgent
Escalation	None unless explicitly stated
Resolution / supersession	Resolution notice informational; prior unresolved security attention clears.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

37. SECURITY NOTIFICATION CONTENT

- Passwords must never appear in notification previews.
- API keys must never appear.
- Secrets must never appear.
- Bearer tokens must never appear.
- Credential values must never appear.
- Private keys must never appear.
- Unnecessary exploit details must never appear.
- Full sensitive findings must remain in authorised records.
- Unrestricted Client security information must never leak through notifications.

38. NOTIFICATION FAMILY 9 - GOVERNANCE / SYSTEM

Primarily for Super Admin and explicitly authorised governance responsibilities.

GS-01 - GOVERNANCE_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Material governance decision/action is required.
Recipients	Super Admin; explicitly authorised governance user if such responsibility is delegated.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT unless event becomes critical
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Governance action completed/cancelled/superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

GS-02 - PRIVILEGED_ACCESS_OVERSIGHT_REQUIRED

Classification	NOTIFY
Trigger	Privileged access state requires governance review/decision.
Recipients	Super Admin / authorised governance oversight.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Oversight decision completed or state superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	Required where privileged action exists.

GS-03 - ADMINISTRATIVE_CONFIGURATION_OVERSIGHT_REQUIRED

Classification	NOTIFY
Trigger	Significant administrative configuration requires oversight.
Recipients	Super Admin / authorised governance user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	AWARENESS / ACTION_REQUIRED
Priority	P2/P3
In-app	Yes
Email / external delivery	Context dependent
Quiet hours	RESPECT
Digest	YES for non-urgent oversight
Escalation	None unless explicitly stated
Resolution / supersession	Oversight completed or configuration superseded.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable
Implementation notes	Routine administration belongs in Audit Log, not Notification Center.

GS-04 - SYSTEM_GOVERNANCE_ISSUE

Classification	NOTIFY
Trigger	Material platform-level governance issue requires action.
Recipients	Super Admin + specifically authorised responsible users.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P2/P1 depending on impact
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; otherwise RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Issue formally resolved/closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

GS-05 - AUDIT_INTEGRITY_ACTION_REQUIRED

Classification	NOTIFY
Trigger	Audit integrity/tamper-evidence control reports a material state requiring authorised attention.
Recipients	Super Admin + appropriate authorised security/governance user.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	ACTION_REQUIRED
Priority	P1/P2
In-app	Yes
Email / external delivery	IMMEDIATE
Quiet hours	P1 BYPASS; P2 RESPECT
Digest	NO
Escalation	None unless explicitly stated
Resolution / supersession	Audit integrity issue resolved or authorised exception closed.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	This event itself must be traceable.
Implementation notes	This event does not define IO SKY's separate final Audit Log integrity architecture.

GS-06 - GOVERNANCE_INCIDENT_RESOLVED

Classification	NOTIFY
Trigger	Governance/system issue is formally resolved.
Recipients	Authorised stakeholders.
Permission / scope / assignment	Effective permission/scope/assignment as specified
Environment	Where relevant
Client-facing	No unless explicitly stated
Action type	RESOLUTION_NOTICE
Priority	P3
In-app	Yes
Email / external delivery	Context dependent
Quiet hours	RESPECT
Digest	YES if non-urgent
Escalation	None unless explicitly stated
Resolution / supersession	Resolution notice informational; related actionable governance notification clears.
Deduplication	Event ID + resource + recipient + lifecycle occurrence
Deep link	Relevant authorised resource
Sensitive content	Minimal context; no secrets or unauthorised details
Audit relation	As applicable

39. ADMIN PERMISSION PRESETS

The ten existing Admin Permission Presets remain Client Manager, Project Manager, Discovery Call Manager, AI Scan Manager, AI Scan Reviewer, Billing Manager, Communications Manager, Delivery Manager, Operations Manager and Administrative Manager.

These are presets, not additional platform roles. Notification routing follows effective final permissions and assignments. Removing a permission must stop future corresponding notifications.

40. ROLE ROUTING SUMMARY

Super Admin. Receives broad authorised oversight, still subject to anti-noise design. Super Admin does not need every healthy background event.

Admin. Admin role by itself is insufficient. Routing follows Permission + Scope + Assignment + Resource.

Developer. Receives only authorised assigned development/review events. Developer role does not automatically grant Billing, Security, AI Scan Review, AI Scan Publish, Governance or unrelated Client/Project notifications. A Developer may be specifically assigned as AI Scan Reviewer.

Technical Operator. Notifications follow explicit technical scopes: Frontend, Backend, Full Stack, UI/UX, Infrastructure/DevOps, Security and Database/Data. Environment and Project assignment still apply.

Client. Receives only Client-facing events belonging to their own account/resources. Never receives internal Scan analysis, unpublished findings, internal review, Developer activity, Technical Operator

activity, internal Security Center data, governance information or another Client's information.

41. ENVIRONMENT-AWARE TECHNICAL EVENTS

Development, Staging and Production are distinct notification scopes where applicable. A user with Staging access only must not receive Production-sensitive notification content. Production notification deep links require Production authorization.

42. QUIET-HOUR SUMMARY

BYPASS is normally permitted/required for SECURITY_ALERT_CRITICAL, P1 SUSPICIOUS_ACCESS_EVENT, PLATFORM_INCIDENT_CRITICAL, P1 AI_SERVICE_UNAVAILABLE, P1 AI_OPERATION_FAILED, P1 AI_SCAN_PUBLICATION_FAILED, P1 SYSTEM_GOVERNANCE_ISSUE, P1 AUDIT_INTEGRITY_ACTION_REQUIRED and other genuine P1 incidents requiring immediate human intervention.

RESPECT quiet hours for ordinary Discovery Call reminders, project deadline reminders, Client information requests, Billing reminders, Scan review notifications, normal integration failures and ordinary administrative events. The in-app notification may still exist immediately.

43. DIGEST SUMMARY

Events must not be digested when they are P1, imminent deadline reminders, AI Scan Ready for Review, AI Scan Review Approved / Ready for Publication, Publication Failed, Payment Failed, Payment Due Today / Overdue, Project Deadline 24H, active Project Blocker, critical/security action, Discovery Call 1H, or a concrete assigned action.

Low-urgency P3/P4 informational events may be digest eligible.

44. ESCALATION FRAMEWORK

Escalation must only exist where unresolved state genuinely warrants broader attention. It must be lifecycle-aware, deadline-aware, assignment-aware, deduplicated and proportional.

Approved fixed schedules: Discovery Call = 24h preparation + 1h imminent reminder. Project deadline = 7 days + 3 days + 24 hours. Payment = 7 days + 1 day + due date + overdue. AI Scan = immediate Ready for Review followed by deadline-aware reminder/risk/overdue behavior based on the actual configured Scan delivery deadline.

For generic operational blockers/incidents, escalation thresholds may be configurable so developers do not hard-code arbitrary universal periods IO SKY never approved.

45. CLIENT LANGUAGE AND AI SCAN PUBLICATION

When a Client has selected the report language, that language must be validated at Publish time.

Mandatory sequence: Analysis Complete → Ready for Review → Human Review → Approve → Publisher Notification → Publisher opens Approved Report → Publish → Backend validation → Correct report version becomes available in correct Client Portal → Correct Client report language → Client notification.

No AI output may bypass this flow.

46. AUDIT LOG RELATIONSHIP

Notification Center and Audit Log remain separate. Actions such as AI Scan approval, AI Scan publication, privileged access change, significant permission change, sensitive configuration change and security incident handling may generate both notification behavior and Audit Log evidence. That does not merge the two systems.

47. NOTIFICATION TEMPLATE REQUIREMENT

Every catalogue Event ID must have stable template keys. Recommended logical pattern: notification...title, notification...body and notification...action. The development team may choose the actual localisation file structure. Hardcoding one language throughout event logic is not acceptable where translation support is part of the platform.

48. NOTIFICATION PREVIEW DATA MINIMISATION

Rows should contain only enough context to answer: What happened? What does it relate to? When? Do I need to act? Where do I go? Full details remain in the underlying authorised record.

49. DEEP LINK SECURITY

A notification deep link must never be treated as an authorization token. Every destination must re-check current user, current permission, current scope, current assignment, current resource access, current environment and current lifecycle state.

50. DELIVERY FAILURE VERSUS BUSINESS FAILURE

Developers must distinguish a business event failure such as AI_SCAN_PUBLICATION_FAILED from a notification channel delivery failure such as an email provider temporarily failing to deliver AI_SCAN_READY_FOR_REVIEW. They are different concepts. A delivery retry does not create a new business event.

51. REQUIRED MASTER EVENT MATRIX

The implementation handoff must include a developer-readable master matrix containing the following fields for every event. Nothing material should be left only implicit in prose.

Required Master Matrix Columns

Required column
Catalogue reference
Event ID
Family
Trigger
Classification
Recipient(s)
Permission requirement
Scope requirement
Assignment requirement
Environment requirement
Client-facing yes/no
Action type
Priority
In-app
Email
Quiet-hours behavior
Digest eligibility
Escalation behavior
Resolution/supersession condition
Dedupe identity
Deep-link destination
Sensitive-content restriction
Audit relation
Translation/template key

52. CORE ACCEPTANCE TESTS

The Event Catalogue is not implemented correctly unless QA proves at minimum all of the following.

- 1. Client A cannot receive Client B notifications.
- 2. Admin without Billing permission receives no Billing notifications.
- 3. Removing Billing permission stops subsequent Billing visibility.
- 4. Old notification link does not bypass revoked access.
- 5. Ordinary Developer does not receive AI Scan Review notification.
- 6. Explicitly assigned authorised Developer Reviewer receives the assigned Scan Review notification.
- 7. Developer Reviewer cannot Publish merely because they reviewed.
- 8. Admin Reviewer without scan.publish cannot Publish.
- 9. Admin with scan.publish but no review permission cannot approve a review merely through Publish authority.

- 10. Super Admin can Publish an approved Scan.
- 11. Authorised Admin Publisher can Publish when explicitly permitted.
- 12. Reviewer pressing Approve generates publisher attention.
- 13. Client receives no internal Scan review data.
- 14. Client receives no report before successful publication.
- 15. Failed publication generates no false Client availability notification.
- 16. Successful publication places correct version in correct Client Portal.
- 17. Correct Client-selected report language is used.
- 18. One repeated Publish request cannot create duplicate Client publications.
- 19. Discovery Call reschedule cancels old reminders.
- 20. Discovery Call cancellation cancels future reminders.
- 21. Discovery Call 24h reminder occurs only for active current schedule.
- 22. Discovery Call 1h reminder occurs only for active current schedule.
- 23. Completed Project does not receive subsequent deadline reminders.
- 24. Project deadline reminder sequence works at 7d, 3d and 24h.
- 25. Payment completed before due date prevents later due/overdue reminders.
- 26. Payment failed does not expose processor secrets.
- 27. Security event routing requires Security authorization.
- 28. Technical Operator without Security scope cannot access Security Center event.
- 29. Staging-only Technical Operator cannot access Production event.
- 30. P1 critical Security event can bypass quiet hours.
- 31. Normal project reminder respects quiet hours.
- 32. In-app event can still exist during quiet hours.
- 33. Digest does not delay urgent event.
- 34. Delivery retry does not create duplicate logical event.
- 35. Read notification can remain actionable.
- 36. Resolved notification can remain historically visible without appearing actionable.
- 37. Assignment change redirects future notification routing.
- 38. Cross-Project leakage is prevented.
- 39. Cross-environment leakage is prevented.
- 40. Secrets/tokens/passwords never appear in notification previews.
- 41. Routine successful background jobs do not create notification noise.
- 42. Audit Log activity does not automatically create notifications.
- 43. User preferences cannot disable mandatory security/governance delivery.
- 44. User preferences cannot change authorization.
- 45. Event translation does not change Event ID or semantics.
- 46. DST/timezone transitions do not duplicate scheduled reminders.
- 47. Duplicate backend event processing is idempotent.
- 48. Persistent email delivery failures remain observable.
- 49. Publication Approval and Publish are separately auditable.
- 50. Notification deep links always perform current backend authorization.

53. DEFINITION OF DONE - EVENT CATALOGUE

Notification work is not complete merely because events exist. It is complete only when all approved Event IDs exist; routing matches current authorization; assignments dynamically affect recipients; tenant and environment isolation are verified; action/read states are separate; resolution works; reminders work; quiet hours work; digest rules work; escalation works; external retries do not duplicate events; translations use stable keys; delivery analytics are available; Scan review/publish separation is enforced; Client Scan publication is verified; security content is minimised; and acceptance tests pass.

54. DEVELOPMENT TEAM IMPLEMENTATION FREEDOM

The development team remains responsible for choosing the appropriate technical implementation for event bus, queueing technology, scheduler, workers, DB design, idempotency primitives, external delivery provider, retry/backoff mechanism, cache strategy, job processing, observability, localisation library and internal event architecture.

Those choices may not weaken the functional behavior in this document.

55. FINAL FUNCTIONAL RULE

The system must always achieve the right event → to the right authorised person → for the right resource → at the right time → with the right action → through the appropriate channel → without exposing information they are not authorised to know.

The Notification Center must support Awareness → Preparation → Action → Resolution while remaining calm, compact and operationally useful.

APPENDIX A - CONSOLIDATED MASTER EVENT MATRIX

This appendix consolidates every normative event into a single developer-readable registry. The detailed event definitions above remain authoritative where a cell is necessarily abbreviated.

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
DC-01	DISCOVERY_CALL_REQUESTED	Discovery Calls	ACTION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO
DC-02	DISCOVERY_CALL_SCHEDULED	Discovery Calls	PREPARATION_REQUIRED	P3	Yes	MANDATORY	RESPECT	NO
DC-03	DISCOVERY_CALL_SCHEDULED	Discovery Calls	AWARENESS / PREPARATION_REQUIRED	P2 when close to original /new time; otherwise P3	Yes	IMMEDIATE	RESPECT	NO
DC-04	DISCOVERY_CALL_CANCELLED	Discovery Calls	AWARENESS	P3	Yes	IMMEDIATE	RESPECT	NO
DC-05	DISCOVERY_CALL_UPCOMING_24H	Discovery Calls	PREPARATION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO
DC-06	DISCOVERY_CALL_UPCOMING_1H	Discovery Calls	PREPARATION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
DC-07	DISCOVERY_CALL_RETRY_DUE	Discovery Calls	ACTION_REQUIRED	P2/P3 based on urgency	Yes	IMMEDIATE for P2; OPTIONAL for P3	RESPECT	NO
DC-08	DISCOVERY_CALL_FOLLOW_UP_DUE	Discovery Calls	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE	RESPECT	NO
DC-09	DISCOVERY_CALL_INFORMATION_UPDATED	Discovery Calls	AWARENESS / PREPARATION_REQUIRED	P3	Yes	OPTIONAL	RESPECT	YES
DC-10	DISCOVERY_CALL_ACTION_REQUIRED	Discovery Calls	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where time-sensitive	RESPECT	NO
AS-01	AI_SCAN_CREATED	Ai Scans	AWARENESS	P3	Yes	OPTIONAL internally; Client purchase confirmation transactional as appropriate	RESPECT	YES internally where no action is required
AS-02	AI_SCAN_INFORMATION_REQUIRED	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-03	AI_SCAN_SUBMITTED	Ai Scans	AWARENESS	P3	Yes	Client confirmation MANDATORY/transactional; internal OPTIONAL	RESPECT	YES internally where no immediate action is required
AS-04	AI_SCAN_ANALYSIS_EXCEPTION	Ai Scans	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT unless business-critical	NO

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
AS-05	AI_SCAN_READY_FOR_REVIEW	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-06	AI_SCAN_REVIEW_AS_SIGNED	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-07	AI_SCAN_REVIEW_REMINDER	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-08	AI_SCAN_CHANGES_REQUIRED	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-09	AI_SCAN_REVIEW_APPROVED	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-10	AI_SCAN_PUBLICATION_FAILED	Ai Scans	ACTION_REQUIRED	P1/P2 depending on delivery impact	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
AS-11	AI_SCAN_PUBLISHED	Ai Scans	AWARENESS / View report	P3	Yes	IMMEDIATE to Client	RESPECT	NO
AS-12	AI_SCAN_DELIVERY_AT_RISK	Ai Scans	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AS-13	AI_SCAN_DELIVERY_OVERDUE	Ai Scans	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS only if severity warrants; otherwise RESPECT	NO
CA-01	CLIENT_ACCOUNT_CREATED	Clients & Accounts	AWARENESS / ACTION_REQUIRED depending on onboarding	P3	Yes	OPTIONAL internally; onboarding communication as required externally	RESPECT	YES internally if not action-required
CA-02	CLIENT_INFORMATION_REQUIRED	Clients & Accounts	ACTION_REQUIRED for Client	P2/P3	Yes	IMMEDIATE to Client	RESPECT	NO
CA-03	CLIENT_INFORMATION_SUBMITTED	Clients & Accounts	ACTION_REQUIRED or AWARENESS depending on workflow	P3	Yes	OPTIONAL; IMMEDIATE where work is blocked awaiting this input	RESPECT	YES only if no action is urgent
CA-04	CLIENT_MATERIAL_INFORMATION_UPDATED	Clients & Accounts	AWARENESS	P3	Yes	OPTIONAL	RESPECT	YES
CA-05	CLIENT_ACCOUNT_ACTION_REQUIRED	Clients & Accounts	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where deadline exists	RESPECT	NO when action deadline exists
CA-06	CLIENT_ACCOUNT_SECURITY_ACTION_REQUIRED	Clients & Accounts	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; otherwise RESPECT	NO
PD-01	PROJECT_ASSIGNED	Projects & Delivery	AWARENESS / ACTION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO
PD-02	PROJECT_REASSIGNED	Projects & Delivery	AWARENESS	P3	Yes	IMMEDIATE	RESPECT	NO

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
PD-03	PROJECT_MILESTONE_UPDATED	Projects & Delivery	AWARENESS / PREPARATION_REQUIRED	P3	Yes	OPTIONAL	RESPECT	YES if not time-critical
PD-04	PROJECT_DEADLINE_7D	Projects & Delivery	PREPARATION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO
PD-05	PROJECT_DEADLINE_3D	Projects & Delivery	ACTION_REQUIRED / PREPARATION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PD-06	PROJECT_DEADLINE_24H	Projects & Delivery	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PD-07	PROJECT_DEADLINE_MISSED	Projects & Delivery	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PD-08	PROJECT_REVIEW_REQUESTED	Projects & Delivery	ACTION_REQUIRED	P2/P3 depending on deadline	Yes	IMMEDIATE	RESPECT	NO
PD-09	PROJECT_REVIEWER_ASSIGNED	Projects & Delivery	ACTION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO
PD-10	PROJECT_IMPLEMENTATION_ACTION_REQUIRED	Projects & Delivery	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where deadline/action is material	RESPECT	NO
PD-11	PROJECT_REVIEW_ACTION_REQUIRED	Projects & Delivery	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PD-12	PROJECT_BLOCKER_CREATED	Projects & Delivery	ACTION_REQUIRED	P2; P1 only if business-critical	Yes	IMMEDIATE	P1 BYPASS; otherwise RESPECT	NO
PD-13	PROJECT_BLOCKER_RESOLVED	Projects & Delivery	RESOLUTION_NOTICE	P3	Yes	OPTIONAL	RESPECT	YES if non-urgent
PD-14	PROJECT_DEPENDENCY_ACTION_REQUIRED	Projects & Delivery	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where delivery is affected	RESPECT	NO
PD-15	PROJECT_CLIENT_INFORMATION_REQUIRED	Projects & Delivery	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE to Client	RESPECT	NO
PD-16	PROJECT_CLIENT_INFORMATION_RECEIVED	Projects & Delivery	AWARENESS / ACTION_REQUIRED	P3	Yes	OPTIONAL; IMMEDIATE where work was blocked	RESPECT	YES only if no immediate action is required
PD-17	PROJECT_DELIVERY_STATE_CHANGED	Projects & Delivery	AWARENESS / ACTION_REQUIRED depending on state	P3 or P2	Yes	Context dependent	RESPECT	YES only for non-urgent informational transitions
PB-01	INVOICE_CREATED_ACTION_REQUIRED	Payments & Billing	ACTION_REQUIRED	P3	Yes	OPTIONAL	RESPECT	YES
PB-02	PAYMENT_DUE_7D	Payments & Billing	PREPARATION_REQUIRED	P3	Yes	IMMEDIATE	RESPECT	NO

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
PB-03	PAYMENT_DUE_1D	Payments & Billing	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE	RESPECT	NO
PB-04	PAYMENT_DUE_TODAY	Payments & Billing	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PB-05	PAYMENT_OVERDUE	Payments & Billing	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PB-06	PAYMENT_FAILED	Payments & Billing	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PB-07	PAYMENT_COMPLETE_D	Payments & Billing	AWARENESS	P4/P3	Yes	Client receipt/transactional as applicable; internal OPTIONAL	RESPECT	YES internally
PB-08	BILLING_ACTION_REQUIRED	Payments & Billing	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where time-sensitive	RESPECT	NO
AO-01	AI_OPERATION_FAILED	Ai Operations	ACTION_REQUIRED	P2/P1 depending on impact	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
AO-02	AI_HUMAN_INTERVENTION_REQUIRED	Ai Operations	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AO-03	AI_PROCESSING_EXCEPTION	Ai Operations	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
AO-04	AI_OPERATION_THRESHOLD_EXCEEDED	Ai Operations	ACTION_REQUIRED	P2/P3 depending on threshold	Yes	IMMEDIATE where action required	RESPECT	NO if action required
AO-05	AI_SERVICE_UNAVAILABLE	Ai Operations	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
AO-06	AI_OPERATION_RECOVERED	Ai Operations	RESOLUTION_NOTICE	P3	Yes	OPTIONAL	RESPECT	YES where no immediate resumed action is required
PI-01	INTEGRATION_DISCONNECTED	Platform & Integrations	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT unless critical Production impact	NO
PI-02	INTEGRATION_AUTH_EXPIRED	Platform & Integrations	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
PI-03	INTEGRATION_FAILURE	Platform & Integrations	ACTION_REQUIRED	P2/P1	Yes	IMMEDIATE	P1 BYPASS; otherwise RESPECT	NO
PI-04	EXTERNAL_SERVICE_UNAVAILABLE	Platform & Integrations	ACTION_REQUIRED	Impact-based P1/P2/P3	Yes	IMMEDIATE for P1/P2	P1 BYPASS; otherwise RESPECT	NO when impact is active
PI-05	PLATFORM_COMPONENT_ACTION_REQUIRED	Platform & Integrations	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT unless separately P1	NO

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
PI-06	PLATFORM_CONFIGURATION_ACTION_REQUIRED	Platform & Integrations	ACTION_REQUIRED	P2/P3	Yes	IMMEDIATE where blocking	RESPECT	NO
PI-07	PLATFORM_INCIDENT_CRITICAL	Platform & Integrations	ACTION_REQUIRED	P1	Yes	IMMEDIATE	BYPASS	NEVER
PI-08	PLATFORM_SERVICE_RECOVERED	Platform & Integrations	RESOLUTION_NOTICE	P3	Yes	OPTIONAL / IMMEDIATE if teams were actively responding	RESPECT	YES where no immediate restart action is needed
SA-01	SECURITY_ALERT_CRITICAL	Security & Access	ACTION_REQUIRED	P1	Yes	IMMEDIATE / MANDATORY	BYPASS	NEVER
SA-02	SECURITY_ACTION_REQUIRED	Security & Access	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT unless severity escalates	NO
SA-03	PRIVILEGED_ACCESS_GRANTED	Security & Access	AWARENESS / oversight	P2/P3	Yes	IMMEDIATE for sensitive privilege	RESPECT	NO for high privilege
SA-04	PRIVILEGED_ACCESS_REVOKED	Security & Access	AWARENESS	P2/P3	Yes	Context dependent	RESPECT	Context dependent
SA-05	SIGNIFICANT_PERMISSION_CHANGE	Security & Access	AWARENESS / ACTION_REQUIRED where review is needed	P2/P3	Yes	IMMEDIATE where high privilege	RESPECT	Context dependent
SA-06	SECURITY_CENTER_ACTION_REQUIRED	Security & Access	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT unless escalated to P1	NO
SA-07	SECURITY_CONFIGURATION_ACTION_REQUIRED	Security & Access	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
SA-08	SUSPICIOUS_ACCESS_EVENT	Security & Access	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
SA-09	ACCOUNT_SECURITY_ACTION_REQUIRED	Security & Access	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
SA-10	SECURITY_INCIDENT_RESOLVED	Security & Access	RESOLUTION_NOTICE	P3	Yes	IMMEDIATE where incident response was active	RESPECT	YES if non-urgent
GS-01	GOVERNANCE_ACTION_REQUIRED	Governance / System	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT unless event becomes critical	NO
GS-02	PRIVILEGED_ACCESS_OVERSIGHT_REQUIRED	Governance / System	ACTION_REQUIRED	P2	Yes	IMMEDIATE	RESPECT	NO
GS-03	ADMINISTRATIVE_CONFIGURATION_OVERSIGHT_REQUIRED	Governance / System	AWARENESS / ACTION_REQUIRED	P2/P3	Yes	Context dependent	RESPECT	YES for non-urgent oversight

Ref	Event ID	Family	Action	Priority	In-app	Email	Quiet	Digest
GS-04	SYSTEM_GOVERNANCE_ISSUE	Governance / System	ACTION_REQUIRED	P2/P1 depending on impact	Yes	IMMEDIATE	P1 BYPASS; otherwise RESPECT	NO
GS-05	AUDIT_INTEGRITY_ACTION_REQUIRED	Governance / System	ACTION_REQUIRED	P1/P2	Yes	IMMEDIATE	P1 BYPASS; P2 RESPECT	NO
GS-06	GOVERNANCE_INCIDENT_RESOLVED	Governance / System	RESOLUTION_NOTICE	P3	Yes	Context dependent	RESPECT	YES if non-urgent

APPENDIX A.2 - ROUTING / LIFECYCLE MATRIX

Ref	Recipients	Client	Escalation	Resolution	Deliver
DC-01	Super Admin; Admin with effective Discovery Call permission and relevant responsibility/assignment.	Unauthenticated prospect does not require an in-app object; external confirmation uses approved scheduling/transactional channel.	If request remains unassigned/unhandled beyond an approved configurable operational threshold.	Request handled, scheduled, cancelled or otherwise closed.	Discovery Call request ID + recipient
DC-02	Current assigned responsible internal user; authenticated Client where applicable; external prospect via approved transactional scheduling channel.	Yes where authenticated; external prospect via approved transactional channel.	None unless explicitly stated	Call cancelled, completed or rescheduled.	Call ID + scheduled occurrence + recipient
DC-03	Current assigned responsible user(s) and relevant authenticated/external participant.	Yes where relevant.	None unless explicitly stated	New schedule becomes authoritative.	Call ID + reschedule version + recipient
DC-04	Current responsible users and relevant participant.	Yes where relevant.	None unless explicitly stated	Cancellation is final unless Call is explicitly reopened/rebooked.	Event ID + resource + recipient + lifecycle occurrence
DC-05	Current assigned authorised internal responsible user.	No unless explicitly stated	None unless explicitly stated	Call cancelled, rescheduled, completed, or 1h reminder becomes the current imminent reminder.	Call ID + scheduled occurrence + 24H + recipient
DC-06	Current assigned authorised responsible user.	No unless explicitly stated	None unless explicitly stated	Call starts/completes/cancels/reschedules.	Call ID + scheduled occurrence + 1H + recipient
DC-07	Current assigned responsible user.	No unless explicitly stated	Based on configured next-action policy if repeatedly unhandled.	Retry completed, cancelled or superseded by new follow-up.	Call ID + retry timestamp/version + recipient
DC-08	Assigned responsible user.	No unless explicitly stated	None unless explicitly stated	Follow-up completed, cancelled or superseded.	Call ID + follow-up occurrence + recipient
DC-09	Current responsible user(s).	No unless explicitly stated	None unless explicitly stated	Informational; no persistent action state unless underlying action is separately created.	Event ID + resource + recipient + lifecycle occurrence
DC-10	Clearly responsible authorised user(s).	No unless explicitly stated	None unless explicitly stated	Underlying action completed/cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
AS-01	Relevant authorised AI Scan Operations user(s) where operational awareness is required.	Client may receive purchase confirmation through appropriate Client-facing channel.	None unless explicitly stated	Scan progresses or is otherwise closed.	Event ID + resource + recipient + lifecycle occurrence

Ref	Recipients	Client	Escalation	Resolution	Deduping
AS-02	Correct Client; authorised assigned Scan Operations responsible user(s).	Yes - only describe what the Client needs to provide.	If information remains outstanding and delivery window is threatened, escalate internally to responsible Scan Operations + Super Admin.	Required information supplied or requirement explicitly cancelled.	Event ID + resource + recipient + lifecycle occurrence
AS-03	Assigned authorised Scan Operations users; Super Admin where appropriate for oversight.	Yes - submission confirmation.	None unless explicitly stated	Analysis begins/progresses.	Event ID + resource + recipient + lifecycle occurrence
AS-04	Authorised Scan Operations responsible user; relevant authorised technical operator where technically assigned; Super Admin when impact is material.	No internal technical detail.	Escalate to Super Admin when unresolved and Scan delivery becomes at risk.	Analysis successfully resumes/completes or Scan is operationally closed.	Event ID + resource + recipient + lifecycle occurrence
AS-05	Super Admin; Admin explicitly authorised with scan.review and access/assignment; Developer only when explicitly assigned as Reviewer for this specific Scan and given required review authorization.	No. Never Client-visible before authorised publication.	Deadline-aware; AI_SCAN_DELIVERY_AT_RISK when configured delivery window is threatened.	Review completed; reviewer reassigned where old reviewer is no longer responsible; Scan returned to analysis/correction; or Scan cancelled/superseded.	Scan ID + review lifecycle version + recipient
AS-06	Newly assigned authorised Reviewer.	No unless explicitly stated	None unless explicitly stated	Review completed, reassigned or cancelled.	Scan ID + assignment version + reviewer
AS-07	Current assigned Reviewer.	No unless explicitly stated	If delivery risk threshold is reached, generate AI_SCAN_DELIVERY_AT_RISK.	Review completed/reassigned/cancelled/superseded.	Scan ID + reminder threshold + current reviewer
AS-08	Appropriate authorised person/team responsible for correction or analysis.	No.	None unless explicitly stated	Revised version becomes available for review.	Event ID + resource + recipient + lifecycle occurrence
AS-09	Super Admin; Admin with explicit scan.publish permission AND access to that Scan.	No until publication succeeds.	None unless explicitly stated	Successfully published; approval revoked through authorised correction workflow; or Scan superseded.	Scan ID + approved report version + publisher recipient
AS-10	Attempting Publisher; Super Admin.	No notification.	P1 immediate; P2 according to delivery risk.	Successful publication or deliberate authorised operational resolution.	Scan ID + publication failure lifecycle; technical retries must not flood users.
AS-11	Correct Client; relevant authorised internal users where useful.	Yes.	None unless explicitly stated	Informational availability event; report remains available according to portal lifecycle.	Scan ID + published approved version + recipient
AS-12	Current responsible reviewer/publisher as applicable; Super Admin.	No unless explicitly stated	This event is itself an escalation state.	Risk cleared through progression or Scan delivery completed.	Scan ID + risk lifecycle/version + recipient

Ref	Recipients	Client	Escalation	Resolution	Deliverables
AS-13	Super Admin; current responsible authorised internal user(s).	No internal workflow details are exposed.	None unless explicitly stated	Delivery completed or authorised exception resolved.	Event ID + resource + recipient + lifecycle occurrence
CA-01	Assigned authorised Client-management users.	Client may receive onboarding/account-access communication separately.	None unless explicitly stated	Onboarding/assignment completed or account state superseded.	Event ID + resource + recipient + lifecycle occurrence
CA-02	Correct Client; assigned responsible internal user where tracking is useful.	Yes.	None unless explicitly stated	Requested information received or request cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
CA-03	Current assigned responsible internal user(s).	No unless explicitly stated	None unless explicitly stated	Underlying request/workflow progresses or is closed.	Event ID + resource + recipient + lifecycle occurrence
CA-04	Assigned relevant user(s).	No unless explicitly stated	None unless explicitly stated	Informational unless a separate action is created.	Event ID + resource + recipient + lifecycle occurrence
CA-05	Correct Client and/or assigned authorised internal user dictated by underlying action.	No unless explicitly stated	None unless explicitly stated	Underlying action completed/cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
CA-06	Affected Client; authorised security oversight only where necessary.	Yes, but no internal Security Center details.	None unless explicitly stated	Required security action completed or authorised resolution.	Event ID + resource + recipient + lifecycle occurrence
PD-01	Newly assigned authorised user.	No unless explicitly stated	None unless explicitly stated	Assignment completed, removed or reassigned.	Project ID + assignment version + recipient
PD-02	New responsible user; previous user only where operational handoff awareness is necessary.	No unless explicitly stated	None unless explicitly stated	New assignment becomes authoritative.	Event ID + resource + recipient + lifecycle occurrence
PD-03	Assigned authorised Project participants whose work/responsibility is affected.	No unless explicitly stated	None unless explicitly stated	Informational or superseded by later milestone state.	Event ID + resource + recipient + lifecycle occurrence
PD-04	Responsible Implementer/owner + relevant Project/Delivery Manager.	No unless explicitly stated	None unless explicitly stated	Completed, cancelled or superseded.	Project/milestone ID + deadline version + 7D + recipient
PD-05	Current responsible users + appropriate delivery oversight.	No unless explicitly stated	None unless explicitly stated	Completed, cancelled or superseded.	Project/milestone ID + deadline version + 3D + recipient
PD-06	Responsible user + appropriate Project/Delivery oversight.	No unless explicitly stated	None unless explicitly stated	Completed, cancelled or superseded.	Project/milestone ID + deadline version + 24H + recipient

Ref	Recipients	Client	Escalation	Resolution	Deluge
PD-07	Assigned responsible user; Project Manager/Delivery Manager with effective authorization; Super Admin where escalation policy requires.	No unless explicitly stated	Escalate according to authorised Project/Delivery responsibility and configured policy.	Work completed or deadline/status formally superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-08	Explicitly assigned Reviewer.	No unless explicitly stated	None unless explicitly stated	Review completed/re assigned/cancelled.	Event ID + resource + recipient + lifecycle occurrence
PD-09	Newly assigned Reviewer.	No unless explicitly stated	None unless explicitly stated	Review completed/re assigned.	Event ID + resource + recipient + lifecycle occurrence
PD-10	Assigned Implementer.	No unless explicitly stated	None unless explicitly stated	Action completed/cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-11	Assigned Reviewer only.	No unless explicitly stated	None unless explicitly stated	Review action completed/reassigned/superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-12	Responsible assignee; relevant Project/Delivery oversight.	No unless explicitly stated	If unresolved at configured blocker escalation threshold or threatens Project deadline, escalate to higher delivery oversight/Super Admin as authorised.	Blocker marked resolved or superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-13	Users whose next action/dependency changes because of resolution.	No unless explicitly stated	None unless explicitly stated	Resolution notice is informational; related blocker notification must clear.	Event ID + resource + recipient + lifecycle occurrence
PD-14	Assigned dependency owner + relevant Project oversight.	No unless explicitly stated	None unless explicitly stated	Dependency satisfied, cancelled or superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-15	Correct Client; assigned Project responsible user.	Yes.	None unless explicitly stated	Information received or request cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
PD-16	Current assigned internal responsible user.	No unless explicitly stated	None unless explicitly stated	Underlying Project work progresses.	Event ID + resource + recipient + lifecycle occurrence
PD-17	Only authorised stakeholders affected by the transition.	No unless explicitly stated	None unless explicitly stated	Superseded by next authoritative delivery state.	Event ID + resource + recipient + lifecycle occurrence

Ref	Recipient(s)	Client	Escalation	Resolution	Deduping
PB-01	Authorised Billing user.	No unless explicitly stated	None unless explicitly stated	Invoice issue/action resolved.	Event ID + resource + recipient + lifecycle occurrence
PB-02	Correct Client billing recipient; authorised internal Billing responsible user where operationally relevant.	Yes.	None unless explicitly stated	Payment completed, invoice cancelled or superseded.	Invoice ID + due-date version + 7D + recipient
PB-03	Correct Client billing recipient + authorised Billing responsible user as appropriate.	Yes.	None unless explicitly stated	Payment completed, invoice cancelled or superseded.	Invoice ID + due-date version + 1D + recipient
PB-04	Correct Client billing recipient + authorised Billing responsible user.	Yes.	None unless explicitly stated	Payment completed, invoice cancelled or superseded.	Invoice ID + due-date version + DUE + recipient
PB-05	Client billing recipient; authorised internal Billing responsible user.	Yes.	Further commercial/legal action is outside this catalogue and must follow approved IO SKY contractual/legal rules.	Payment completed, invoice legally/operationally superseded or authorised closure.	Event ID + resource + recipient + lifecycle occurrence
PB-06	Correct Client; authorised Billing responsible user.	Yes.	None unless explicitly stated	Successful payment or authorised alternative resolution.	Event ID + resource + recipient + lifecycle occurrence
PB-07	Client confirmation; relevant Billing user where operationally useful.	Yes as appropriate.	None unless explicitly stated	Informational.	Event ID + resource + recipient + lifecycle occurrence
PB-08	Authorised Billing responsible user and/or correct Client where underlying action requires.	No unless explicitly stated	None unless explicitly stated	Underlying action completed/cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
AO-01	Authorised assigned AI/technical responsible users.	No unless explicitly stated	None unless explicitly stated	Process successfully recovered/restarted or intentionally terminated.	Event ID + resource + recipient + lifecycle occurrence
AO-02	Assigned authorised responsible user.	No unless explicitly stated	None unless explicitly stated	Human intervention completed or workflow superseded.	Event ID + resource + recipient + lifecycle occurrence
AO-03	Assigned operational/technical responsible users.	No unless explicitly stated	None unless explicitly stated	Exception resolved/recovered/superseded.	Event ID + resource + recipient + lifecycle occurrence
AO-04	Authorised responsible user(s).	No unless explicitly stated	None unless explicitly stated	Threshold returns to acceptable state or authorised resolution.	Event ID + resource + recipient + lifecycle occurrence

Ref	Recipients	Client	Escalation	Resolution	Redupe
AO-05	Assigned authorised operational/technical users + Super Admin for material outage.	No unless explicitly stated	None unless explicitly stated	Service restored or authorised workaround/closure.	Event ID + resource + recipient + lifecycle occurrence
AO-06	Users actively affected/responsible.	No unless explicitly stated	None unless explicitly stated	Resolution notice is informational; prior incident attention clears.	Event ID + resource + recipient + lifecycle occurrence
PI-01	Authorised owner/operator of affected integration.	No unless explicitly stated	None unless explicitly stated	Integration restored or intentionally disconnected.	Event ID + resource + recipient + lifecycle occurrence
PI-02	Authorised integration owner/operator.	No unless explicitly stated	None unless explicitly stated	Authentication restored or integration intentionally disabled.	Event ID + resource + recipient + lifecycle occurrence
PI-03	Appropriate authorised technical/operational owner.	No unless explicitly stated	None unless explicitly stated	Integration recovers or issue is resolved.	Event ID + resource + recipient + lifecycle occurrence
PI-04	Authorised responsible technical/operational users.	No unless explicitly stated	None unless explicitly stated	Service recovers or authorised workaround/closure.	Event ID + resource + recipient + lifecycle occurrence
PI-05	Authorised component owner/operator.	No unless explicitly stated	None unless explicitly stated	Required intervention completed or issue superseded.	Event ID + resource + recipient + lifecycle occurrence
PI-06	Authorised responsible user.	No unless explicitly stated	None unless explicitly stated	Configuration corrected or issue superseded.	Event ID + resource + recipient + lifecycle occurrence
PI-07	Appropriate authorised technical responsible users; Super Admin.	No unless explicitly stated	Immediate according to authorised operational incident chain.	Incident confirmed resolved.	Event ID + resource + recipient + lifecycle occurrence
PI-08	Affected responsible users.	No unless explicitly stated	None unless explicitly stated	Resolution notice informational; related incident attention clears.	Event ID + resource + recipient + lifecycle occurrence
SA-01	Super Admin; explicitly authorised Security user(s).	No unless explicitly stated	Immediate.	Security incident formally resolved/closed.	Event ID + resource + recipient + lifecycle occurrence
SA-02	Explicitly authorised Security user(s).	No unless explicitly stated	None unless explicitly stated	Security action completed or issue superseded.	Event ID + resource + recipient + lifecycle occurrence

Ref	Recipients	Client	Escalation	Resolution	Deluge
SA-03	Super Admin and authorised governance/security oversight where appropriate.	No unless explicitly stated	None unless explicitly stated	Informational/oversight state completed or superseded.	Event ID + resource + recipient + lifecycle occurrence
SA-04	Appropriate authorised oversight users.	No unless explicitly stated	None unless explicitly stated	Informational.	Event ID + resource + recipient + lifecycle occurrence
SA-05	Authorised oversight.	No unless explicitly stated	None unless explicitly stated	Oversight completed or change superseded.	Event ID + resource + recipient + lifecycle occurrence
SA-06	Only users with explicit Security Center authorization.	No unless explicitly stated	None unless explicitly stated	Security Center action completed/resolved.	Event ID + resource + recipient + lifecycle occurrence
SA-07	Explicit Security-authorised users.	No unless explicitly stated	None unless explicitly stated	Configuration corrected/resolved.	Event ID + resource + recipient + lifecycle occurrence
SA-08	Explicit Security-authorised users; affected user only when appropriate user-facing security action is required.	No unless explicitly stated	None unless explicitly stated	Investigation/action resolved or event dismissed by authorised workflow.	Event ID + resource + recipient + lifecycle occurrence
SA-09	Affected authorised account holder.	May be Client-facing when the affected account is a Client.	None unless explicitly stated	Required account security action completed or authorised resolution.	Event ID + resource + recipient + lifecycle occurrence
SA-10	Authorised stakeholders who were responsible/affected.	No unless explicitly stated	None unless explicitly stated	Resolution notice informational; prior unresolved security attention clears.	Event ID + resource + recipient + lifecycle occurrence
GS-01	Super Admin; explicitly authorised governance user if such responsibility is delegated.	No unless explicitly stated	None unless explicitly stated	Governance action completed/cancelled/superseded.	Event ID + resource + recipient + lifecycle occurrence
GS-02	Super Admin / authorised governance oversight.	No unless explicitly stated	None unless explicitly stated	Oversight decision completed or state superseded.	Event ID + resource + recipient + lifecycle occurrence
GS-03	Super Admin / authorised governance user.	No unless explicitly stated	None unless explicitly stated	Oversight completed or configuration superseded.	Event ID + resource + recipient + lifecycle occurrence
GS-04	Super Admin + specifically authorised responsible users.	No unless explicitly stated	None unless explicitly stated	Issue formally resolved/closed.	Event ID + resource + recipient + lifecycle occurrence

Ref	Recipients	Client	Escalation	Resolution	Dedupe
GS-05	Super Admin + appropriate authorised security/governance user.	No unless explicitly stated	None unless explicitly stated	Audit integrity issue resolved or authorised exception closed.	Event ID + resource + recipient + lifecycle occurrence
GS-06	Authorised stakeholders.	No unless explicitly stated	None unless explicitly stated	Resolution notice informational; related actionable governance notification clears.	Event ID + resource + recipient + lifecycle occurrence